

Active Inference ~ Parr, Pezzulo, Friston ~ Chapter 4 ~ BookStream #002.0

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Chapter four is called the generative models of active inference and it begins with a quotation everything should be made as simple as possible but not simpler by Albert Einstein.

Ali, what is your overview thought or warning for chapter four?

Okay, so after the preliminary materials in chapters two and three, which was basically largely based on providing some conceptual framework for developing the further theory, chapter four delves into much more detail in terms of mathematical formulation and it unpacks a lot more the way that the central equations of active inference is derived and how to construct the important elements of active inference models.

So say matrices A, B, C and D and also how to put together generative models in different situations.

So it basically lays out the foundation for constructing active inference models,

both for discrete time situations and continuous time ones,

which will be used later in chapters seven and eight.

But this is probably one of the most challenging and at least mathematically dense chapters in the book.

So I would personally suggest reading through this chapter really slowly.

And even if we don't get to understand every single detail of the chapter,

obviously we can return per required as we go through the textbook.

Thank you, Ali.

Yes.

So let's look through the sections.

Just to add on though, chapter four is one of the larger and more equation dense chapters

because it is the common kernel or basis that's then going to get applied in chapter five in the neurobiological case.

There's a recipe for making chapter four in chapter six.

That's the recipe for active inference modeling.

Chapter seven and eight are about the discrete and the continuous time variant or subtype or motif of these kinds of things called generative models.

So this is the real common root and we'll just look at what the sections are.

This chapter complements the preceding chapters conceptual treatment of active inference with a more formal treatment.

Section 4.2 from Bayesian inference to free energy.

What would you say about this section, Ali?

Okay.

So as we know, the free energy principle is inspired by previous work on Bayesian inference.

I mean, all the way back to Helmholtz theory about unconscious inference or something to that effect.

I can't remember the exact term, but here, I mean, it provides in a bit more detail how we can derive free

energy principle formalism

using the established Bayesian inference formulation.

And particularly one of the key movements or at least one of the key decisions in through the derivation of free energy principle formulation

is using Jane's inequality principle to derive an upper bound instead of just using the exact values to compute or to achieve required parameters.

So that's basically, in my opinion, the key premise of Section 4.2.

And to see in a bit more detail how we can achieve those upper bounds using Jane's inequality directly from, by using, I mean, manipulations of Bayesian inference, Bayesian statistical formalism.

Thanks.

I'll just add one point from this section.

Broadly, these are the problems of inferring states of the world, perception, and inferring a course of action planning.

So this is, again, referring to the perception and action and everything that happens in between is the internal or the cognitive part of the inference.

But this is like the blanket state cybernetic input-output.

And then let's look at the first equation or how much equations overall or what equations do you think we should highlight?

Okay, so maybe we can, I mean, just as a general comment about these different equations.

Well, each of these equations provide a distinct step toward deriving the ultimate whole picture.

So even if we don't quite understand how we can derive from each step to the other one, it's good to know that it's only required to understand how we get to that ultimate whole picture.

But ultimately what we would need in order to develop active inference models is the ultimate equation or ultimate whole picture.

So this is just a way to elucidate the steps toward developing that whole picture.

But again, it's not an essential requirement to understand the materials of the rest of the book.

But if we go from, I mean, equations 4.1 toward the 4.4, or other words, the variation of free energy, well, equation 4.1 is just a basic definition of some properties of probabilities in terms of conditional probability and so on.

So equation 4.2 provides the central Jane's inequality principle and how it relates to, I mean, conditional probabilities and, of course, joint probabilities.

And then by using those two properties or those two equations, we ultimately get to 4.4, which is the definition of variation free energy parameter, which is the parameter of interest that needs to be optimized in order to inference to happen, or at least perceptual inference to happen in active inference models.

Thanks. Only thing I'll add is F is the letter used for variational free energy.

You can think of it like a computer program,

and the arguments that it takes in or the variables that it takes in are Q ,

which is the distribution that's under the statistician's control,

and Y , which are the data which are outside of the statistician's control.

And do you want to describe more about anything in this equation or carry on?

Just one thing that can probably be helpful is to somehow compare these steps with the initial picture we had from chapter 2,

because variation free energy was first introduced in chapter 2.

So it can be helpful to go back and forth between chapters 2 and 4

and try to connect the dots between the related points there.

Section 4.3, Generative Models.

All right, I'll read the first sentence, then you can give some thoughts.

To calculate the free energy, we need three things.

Data, a family of variational distributions,

and a generative model comprising a prior and a likelihood.

In this section, we outline two very general sorts of generative model

used for active inference and the form the free energy takes in relation to each.

Okay, so as mentioned earlier,

this chapter deals both with discrete time and continuous time situations.

So clearly, we would need two different types of generative models for each situation.

And obviously, the generative models or the way to construct generative models for discrete time situations would vary quite a bit from the one for continuous time situations.

But the general principle underlying those generative models are basically the same, which is, I mean, to somehow construct a model of the environment.

I mean, either be it for the situation that is sequential in time,

or for the situations that need to be somehow,

each moment of the situation needs to be accommodated in terms of a continuous time situation.

So figure 4.2 provides some examples of both.

So for, yes, let me see.

Yes, so we have some examples of different kinds of generative models, some case studies, if you like.

And it provides various ways to show how the dependencies between variables can be modeled using these kinds of graphical probabilistic models.

So one common way to represent generative models is to use these kinds of graphical probabilistic models in active inference literature, which is, at least in this case,

the circles would represent the random variables,

and the squares would represent the distributions,

which would describe the dependencies between those random variables.

So we can see the clear relationships between those parameters here,

which is basically what this whole graph,

what constitutes the generative model that needs to be used for different situations.

And then in figure 4.3,
we can compare the two different types of generative models
based on whether it's discrete time or continuous time situations.
So the upper picture is a generative model for the discrete time situation,
and the lower picture is the parallel continuous time version of it.
And as we can see, the general topology of these models are the same.
The only things that differ is the use of parameters for policies or,
I mean, discrete time policies or the continuous time ones.
And we can obviously compare the different elements for both prior states
and external states, internal states, and so on,
by comparing these two models here.
Yeah, we often return to figure 4.3.
It's kind of the Rosetta Stone of generative modeling for the context of this book,
because it's then going to develop out into chapter 7 and 8,
and it represents a really fundamental decision made in modeling.
And in the later chapters,
it's also shown how it can be made into a hierarchical model
that combines aspects of both.
But within each level of modeling,
still these are the kinds of decisions that modelers are presented with
when it comes to statistical modeling overall.
So section 4.4 goes into essentially the top half of figure 4.3,
discrete time.
What would you say about discrete time?
Okay, so the discrete time situation
is obviously the archetype discrete time situation,
which is the POMDP models.
So at this point,
I would very much like to recommend
following the steps,
following the material from step-by-step paper,
because that's...
In that paper,
the way to construct POMDP models
is described in a bit more detail.
So if anyone feels like they should
learn a bit more about
the gaps in the details,

I would very much like to
recommend that particular paper.

So, yeah,

I don't know how much detail
we should go into,
because it's...

I mean,

although it's not maybe
detailed enough for some tastes,
but it goes in a quite extensive detail
about how we can construct these models
using the concepts we've learned
in previous chapters.

So,

ultimately,

we reach equations 4.13 and 4.14,
which are basically
the culmination of POMDP formulation
using the vector notations
and gradients and so on.

So...

That's great.

And then we go to continuous time situation.

Yeah.

Great.

A few things intervene
in the continuous time chapter.

Ones that we'll just mention here,
because they're kind of boxed
or partitioned

from the continuous time part,
but they're following pages.

First is Markov blankets.

We won't go into it here,

but kind of footnote that
or look at some other places
where we talk about it
outside of this chapter overview.

Figure 4.4,
Bayesian message passing.
Again,
a big topic.
Let's kind of just go past it now.
Back to the regularly scheduled
continuous time generative model discussion.
and then another box
to the generalized coordinates of motion.
So taking position
plus derivatives of position.
And that has some beneficial properties
that are described
and unpacked also elsewhere.
Do you want to say anything
about 4.5.2?
Well, the only thing
that comes to mind
is although,
as I said before,
all the formulations here
may look more,
I mean,
a bit too dense
to understand
at the first pass.
But some of the key
maybe components here
could be
obviously
the material
from box 4.2
and 4.3
I think
are quite essential
to understand
the underlying principle
behind deriving

the continuous time situation.

Because without

Laplace approximation,

what we would have

in terms of

free energy minimization

is,

it would look

very much like

the Gibbs free energy.

So,

I mean,

the key distinction

between

the free energy principle

as described

in active inference literature

as opposed to

Gibbs free energy

is this distinction,

is the Laplace approximation.

So,

this is what enables us

to go from

Gibbs free energy

to,

I mean,

the variation

of free energy.

So,

yeah,

that's,

I mean,

quite essential

to make this,

to be familiar

with this

essential approximation.

and obviously,
the concept
of generalized
coordinates of motion
will come
time and time again
throughout the whole book,
particularly in chapters
eight and nine.

So,
yeah,
those two concepts
I believe
needs a bit more
attention.

So,
yeah,
sounds good.

Box 4.3,
Laplace approximation,
equations,
another message passing,
representation,
and a summary.

The key message
to take away
is that approximate
Bayesian inference
may be framed
as minimizing
a quantity
known as
variational free energy.

This depends
on a generative model
that expresses
our belief
about how data

are generated.

Anything else

you want to add?

Nothing comes

to mind

at the moment

because,

as I said,

this is,

I mean,

we're still

in the stage

that we want

to develop

our essential tools

to be used

in the rest

of the book.

So,

here,

up until now,

I believe,

by the end

of chapter four,

we have acquired

all the essential,

necessary,

mathematical tools

and the next chapter,

chapter five,

is,

kind of acts

like an interlude.

and I don't think

it's the direct,

I mean,

continuation of chapters

one through four.

So,
I believe
the first section
or the first part
of the book
conceptually
and mathematically
ends here.

So,
yeah,
that's it.

yes,
it's a little bit
like the pragmatic
modeling part
gets foreshadowed
or explored
in five
now that we're
all built up
with four.

All right,
that's the end
of the overview
for four.

five

Thank you.